

RQ Answers

omitted

2024-04-05

Data overview

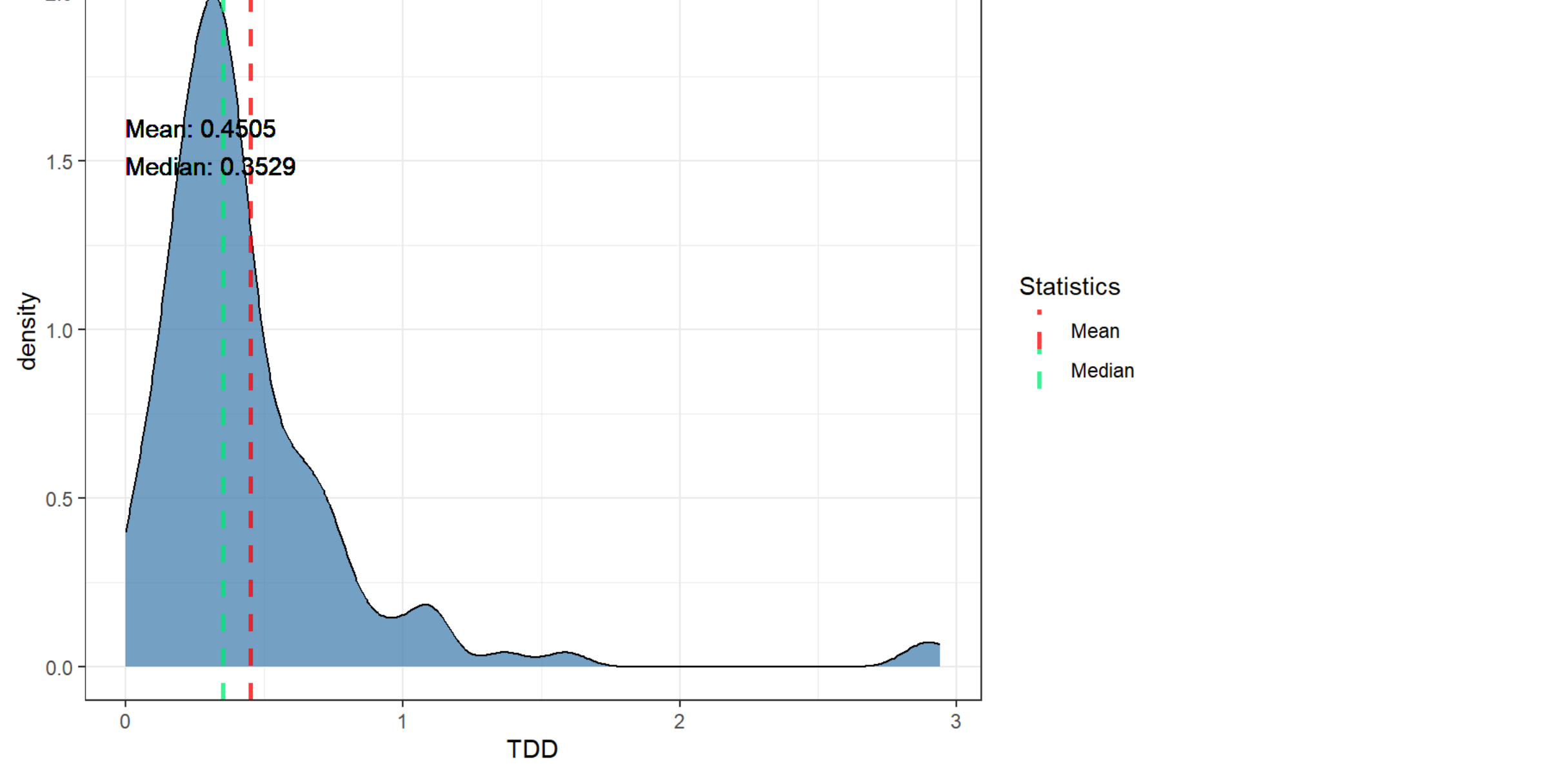
glimpse(all_data)				
## Rows: 5,732				
## Columns: 11				
## \$ repo	<chr> "struts", "struts", "struts", "struts", "struts", ...			
## \$ pr_number	<int> 276, 276, 276, 276, 276, 276, 276, 276, 276, ...			
## \$ rule	<chr> "java:S1133", "java:S1135", "java:S1874", "java:S2...			
## \$ severity	<chr> "INFO", "INFO", "MINOR", "CRITICAL", "MAJOR", "MIN...			
## \$ file	<chr> "core/src/main/java/org/apache/struts2/dispatcher/...			
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...			
## \$ status	<chr> "CLOSED", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPE...			
## \$ debt	<int> 10, 0, 15, 30, 5, 1, 5, 30, 15, 1, 5, 5, 5, 0, ...			
## \$ ncloc_affected_file	<int> 143, 143, 143, 143, 143, 143, 143, 143, 143, ...			
## \$ complexity	<int> 34, 34, 34, 34, 34, 34, 34, 34, 34, 120, 120, ...			
## \$ origin	<chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...			

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 122 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 122 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

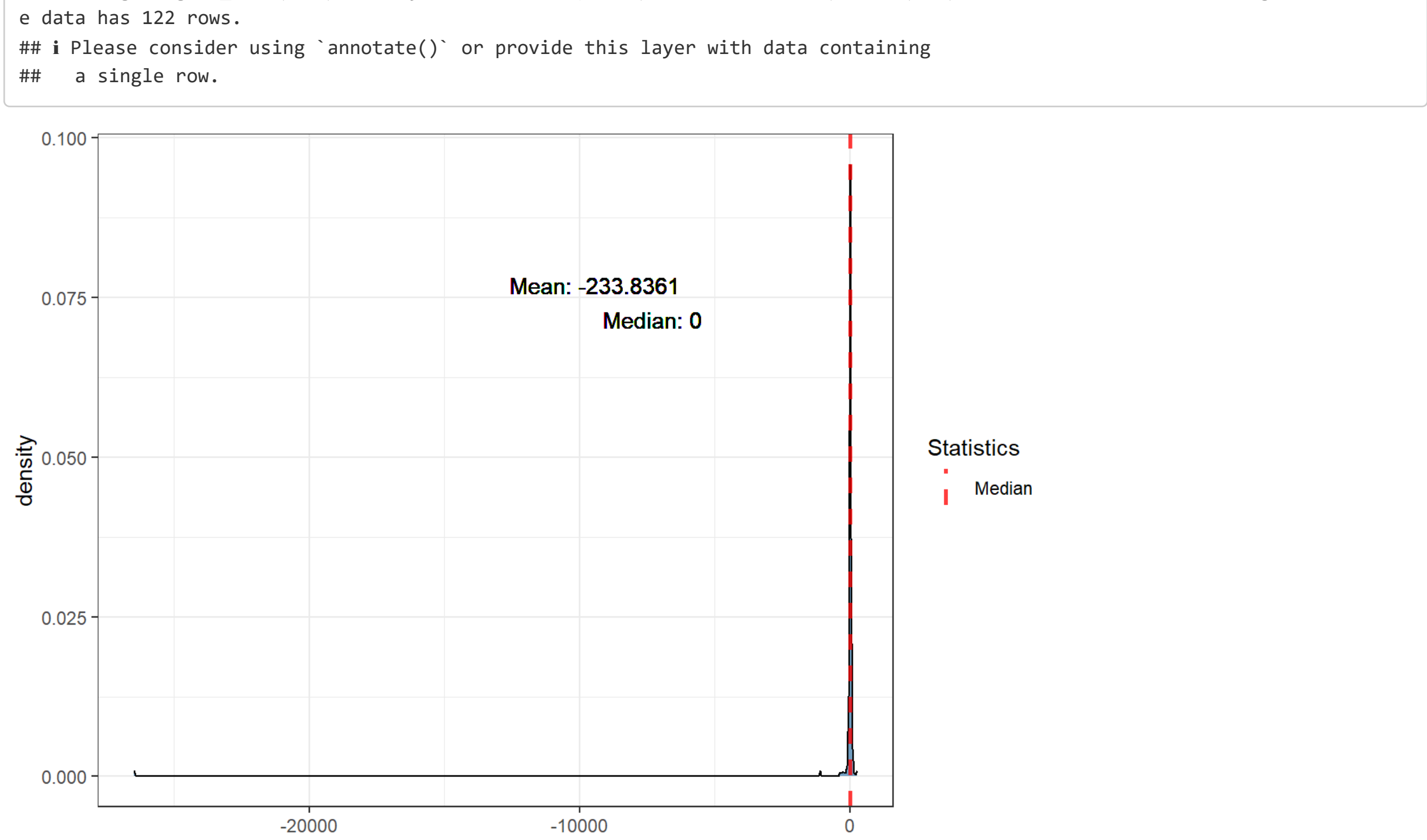


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 122 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 122 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 struts	35.2	44.3	20.5	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	
## 1	35.2	44.3	20.5

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 struts	55.4	44.6	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	55.4	44.6

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 struts	85.7	14.3	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	85.7	14.3

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (*./issues_characterization.Rmd*), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank	struts	
## <int>	<chr>	
## 1	1 java:S3740	- 541
## 2	2 java:S1161	- 313
## 3	3 java:S1192	- 228
## 4	4 java:S1874	- 200
## 5	5 java:S3776	- 178
## 6	6 java:S100	- 156
## 7	7 java:S1149	- 138
## 8	8 java:S1124	- 97
## 9	9 java:S2129	- 91
## 10	10 java:S1199	- 70

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S3740	122	
## 2 java:S1161	123	
## 3 java:S1192	124	
## 4 java:S1874	125	
## 5 java:S3776	126	
## 6 java:S100	127	
## 7 java:S1149	128	
## 8 java:S1124	129	
## 9 java:S2129	130	
## 10 java:S1199	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank	struts	
## <int>	<chr>	
## 1	1 java:S3740	- 421
## 2	2 java:S1133	- 209
## 3	3 java:S3776	- 170
## 4	4 java:S1161	- 143
## 5	5 java:S1192	- 132
## 6	6 java:S1117	- 113
## 7	7 java:S1948	- 102
## 8	8 java:S1186	- 68
## 9	9 java:S1874	- 67
## 10	10 java:S1149	- 62

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S3740	122	
## 2 java:S1133	123	
## 3 java:S3776	124	
## 4 java:S1161	125	
## 5 java:S1192	126	
## 6 java:S1117	127	
## 7 java:S1948	128	
## 8 java:S1186	129	
## 9 java:S1874	130	
## 10 java:S1149	131	

Outliers

Higher TDD

## # A tibble: 122 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 struts	670	1950	663	2.94
## 2 struts	811	20	7	2.86
## 3 struts	479	240	151	1.59
## 4 struts	621	894	654	1.37
## 5 struts	298	237	210	1.13
## 6 struts	302	295	270	1.09
## 7 struts	844	280	257	1.09
## 8 struts	671	150	138	1.09
## 9 struts	580	30	31	0.968
## 10 struts	872	30	32	0.938
## # i 112 more rows				

Higher TDV

## # A tibble: 122 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	427 struts	246	
## 2	715 struts	120	
## 3	550 struts	38	
## 4	532 struts	35	
## 5	313 struts	31	
## 6	493 struts	25	
## 7	679 struts	25	
## 8	479 struts	20	
## 9	867 struts	20	
## 10	307 struts	17	
## # i 112 more rows			

Lower TDV

## # A tibble: 122 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	799 struts	-26485	
## 2	670 struts	-1113	
## 3	861 struts	-350	
## 4	657 struts	-247	
## 5	586 struts	-130	
## 6	671 struts	-80	
## 7	808 struts	-73	
## 8	549 struts	-67	
## 9	621 struts	-56	
## 10	622 struts	-50	
## # i 112 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 121	

Number of java:S1192 issues that affect test code

## n	
## 1 0	

Number of java:S1192 issues that affect not test code

## n	
## 1 360	

Summary NCLOC

## ncloc	
## Min. :	7.0
## 1st Qu.:	134.2
## Median:	327.5
## Mean :	847.1
## 3rd Qu.:	660.2
## Max. :	33995.0